D4 Configuration & Context Data

D1 Normalize Network Data

D5 Enforcement Records  Logs

D2 Analysis Results

D3 Rule / Decision Records

**Network_Segment**

| | | |
|---|---|---|
| PK | segment_id | INT |
| | segment_name | TEXT |
| | description | TEXT |
| | subnet | CIDR |
| | created_at | TIMESTAMPTZ |

**Network_Asset**

| | | |
|---|---|---|
| PK | asset_id | INTEGER |
| | asset_name | TEXT |
| | asset_type | TEXT |
| | ip_address | INET |
| | mac_address | MACADDR |
| FK | segment_id | INT |
| | description | TEXT |
| | created_at | TIMESTAMPTZ |

**Monitoring_Source**

| | | |
|---|---|---|
| PK | source_id | INT |
| | source_name | TEXT |
| | source_type | TEXT |
| | collection_methods | TEXT[] |
| | status | TEXT |
| | description | TEXT |

**Enforcement_Target**

| | | |
|---|---|---|
| PK | target_id | INT |
| | target_name | TEXT |
| | target_type | TEXT |
| | ip_address | INET |
| | status | TEXT |
| | created_at | TIMESTAMPTZ |

**User_Information**

| | | |
|---|---|---|
| PK | user_id | INT |
| | username | TEXT |
| | fullname | TEXT |
| | role | TEXT |
| | email | TEXT |
| | status | TEXT |

**Telemetry_Source_Map**

| | | |
|---|---|---|
| PK | source_map_id | BIGINT |
| FK | telemetry_id | BIGINT |
| FK | source_id | INT |
| | source_record_id | TEXT |
| | source_record_type | TEXT |
| | observed_at | TIMESTAMPTZ |

**Telemetry_Record**

| | | |
|---|---|---|
| PK | telemetry_id | BIGINT |
| | record_type | TEXT |
| | event_time | TIMESTAMPTZ |
| | start_time | TIMESTAMPTZ |
| | end_time | TIMESTAMPTZ |
| | src_ip | INET |
| | dst_ip | INET |
| | protocol | TEXT |
| FK | src_asset_id | INT |
| FK | dst_asset_id | INT |
| | create_at | TIMESTAMPTZ |

**Flow_Record**

| | | |
|---|---|---|
| PK FK | telemetry_id | BIGINT |
| | duration | DOUBLE PRECISION |
| | bytes_sent | BIGINT |
| | bytes_received | BIGINT |
| | packets_sent | BIGINT |
| | packets_received | BIGINT |
| | connection_state | TEXT |

**Event_Record**

| | | |
|---|---|---|
| PK FK | telemetry_id | BIGINT |
| | event_type | TEXT |
| | event_catagory | TEXT |
| | event_severity | SMALLINT |
| | additional_data | JSONB |

**Metric_Record**

| | | |
|---|---|---|
| PK FK | telemetry_id | BIGINT |
| | metric_name | TEXT |
| | metric_value | NEMERIC |
| | unit | TEXT |
| | interface_name | TEXT |
| | addtional_data | JSONB |

**PCAP_File**

| | | |
|---|---|---|
| PK | PCAP_id | BIGINT |
| | file_reference | TEXT |
| | capture_start | TIMESTAMPTZ |
| | capture_end | TIMESTAMPTZ |
| | capture_interface | TEXT |
| | capture_source | TEXT |
| | file_source | TEXT |
| | created_at | TIMESTAMPTZ |

**Telemetry_PCAP_Evidence**

| | | |
|---|---|---|
| PK | evidence_id | BIGINT |
| FK | telemetry_id | BIGINT |
| FK | PCAP_id | BIGINT |
| | attribute name | TEXT |

**Enforcement_Record**

| | | |
|---|---|---|
| PK | enforcement_id | BIGINT |
| FK | rule_vesion_id | BIGINT |
| FK | target_id | INT |
| | adapter_type | TEXT |
| | requested_at | TIMESTAMPTZ |
| | executed_at | TIMESTAMPTZ |
| | execution_status | TEXT |
| | response_message | TEXT |
| | created_at | TIMESTAMPTZ |

**Verification_Record**

| | | |
|---|---|---|
| PK | verification_id | BIGINT |
| FK | enforcement_id | BIGINT |
| | verification_method | TEXT |
| | verification_status | TEXT |
| | expected_state | JSONB |
| | observed_state | JSONB |
| | massage | TEXT |
| | veridied_at | TIMESTAMPTZ |

**Analysis_Result**

| | | |
|---|---|---|
| PK | analysis_id | BIGINT |
| | risk_score | NUMERIC |
| | risk_level | TEXT |
| | classification | TEXT |
| | reason | TEXT |
| | model_name | TEXT |
| | model_version | TEXT |
| | analysis_status | TEXT |
| | analysis_at | TIMESTAMPTZ |
| | created_at | TIMESTAMPTZ |

**Analysis_Telemetry**

| | | |
|---|---|---|
| PK | analysis_telemetry_id | BIGINT |
| FK | analysis_id | BIGINT |
| FK | telemetry_id | BIGINT |
| | relation_type | TEXT |
| | created_at | TIMESTAMPTZ |

**Generic_Rule**

| | | |
|---|---|---|
| PK | rule_id | BIGINT |
| FK | decision_id | BIGINT |
| | rule_status | TEXT |
| | create_at | TIMESTAMPTZ |

**Decision_Record**

| | | |
|---|---|---|
| PK | decision_id | BIGINT |
| FK | analysis_id | BIGINT |
| | decision_type | TEXT |
| | reason | TEXT |

**Rule_Version**

| | | |
|---|---|---|
| PK | rule_version_id | BIGINT |
| FK | rule_id | BIGINT |
| | version_no | INT |
| | action | TEXT |
| | src_condition | JSONB |
| | dst_condition | JSONB |
| | protocol | TEXT |
| | port_condition | JSONB |
| | direction | TEXT |
| | priority | INT |
| | status | TEXT |
| | created_at | TIMESTAMPTZ |

**Rule_Decision**

| | | |
|---|---|---|
| PK | rule_decision_id | BIGINT |
| FK | rule_version_id | BIGINT |
| FK | user_id | INT |
| | decision_type | TEXT |
| | comment | TEXT |
| | decided_at | TIMESTAMPTZ |